Spectral truncation of the flat plane: state-space convergence with a squared window, and the disk-with-cone left open

J. Loutey*

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Abstract

The spectral-truncation convergence results of the companion Papers [11, 12] treat compact groups, where a Fejér-type kernel is available and the state spaces are compact. We treat the flat plane $\mathbb{R}^2$ — non-compact, but with a transitive translation group. Truncate $L^2(\mathbb{R}^2)$ to the band-limited subspace $\mathcal{H}_\Lambda$ (Fourier support $|\xi| \le \sqrt{\Lambda}$), compress bounded uniformly continuous multipliers into it, and metrize the truncated state space by the translation Lipschitz seminorm. With a lift built from the translates of one window vector $\psi_\Lambda \in \mathcal{H}_\Lambda$, the compression/lifted-state argument of Paper 38 (whose scalar compact-group form is due to Gaudillot-Estrada and van Suijlekom [1]) goes through with the *positive* kernel $|\psi_\Lambda|^2$. Four estimates hold with the single constant $\gamma_\Lambda = \int |x|\,|\psi_\Lambda(x)|^2 dx = \gamma_1 \Lambda^{-1/2}$ **[INTERNAL THEOREM]**, and they give a Gromov–Hausdorff bound $d_{\mathrm{GH}} \le \gamma_\Lambda$ between the Monge–Kantorovich state spaces, taken over the states of finite first moment **[INTERNAL THEOREM]** for those state classes; whether they are the canonical ones is open. The window must have a finite first moment. The sharp ball window does not: its first moment diverges logarithmically. The smooth-edged windows $(1 - |\xi|^2)_+^m$ with $m \ge 1$ do; for $m = 1$, $\gamma_1 = 32/(5\pi)$. The two-dimensional Bochner–Riesz map is *not* a usable lift: its kernel is negative somewhere at every order, and its Lebesgue constant exceeds 1 **[PRIOR ART]** (both classical facts [10], with a one-line argument recorded here). On the finite Dirichlet disk-with-cone the paper reports measurements. The Markov–Cesàro reconstruction map is not positivity-preserving (a ring counterexample), and its reconstruction error on the open disk decays on every tested observable. Convergence on the disk, and on the cone with apex angle $\alpha \ne 1$ (which has no translation group), is left open.

1 Introduction

1.1 Context

In the spectral-truncation setting of Connes and van Suijlekom [3] a truncated operator system $\mathcal{O}_\Lambda = P_\Lambda \mathcal{A} P_\Lambda$ replaces the algebra of functions, and one asks whether its state space, with a Monge–Kantorovich metric, converges to the state space of the continuum. For compact metric groups Gaudillot-Estrada and van Suijlekom [1] prove this with a compression/lifted-state pair and a single first-moment constant; Paper 38 [11] transports the argument to the spinor window on $S^3 = \mathrm{SU}(2)$, and Paper 40 [12] to compact semisimple Lie groups. All of these carriers are compact.

This paper was first written about the two-dimensional disk-with-cone $\mathbb{D}_{R,\alpha}$, the spatial factor of the near-horizon cigar of the companion gravity paper [13]. It claimed a Bochner–Riesz reconstruction on a "plane $\mathbb{R}^2_\alpha$" and a propinquity backbone for that paper's cigar-tip entropy. Those claims do not hold, and are withdrawn (§1.3). What survives is a flat-plane theorem, proved by a different map, and a set of finite-disk measurements.

*Independent researcher. jloutey@gmail.com. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude. Implementation, internal memos, numerical verification, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility.

1.2 Results

- **(Flat plane, §2.)** For the band-limited truncation of $\mathbb{R}^2$ with the translation seminorm, the compression S and the window lift υ satisfy the four estimates of Paper 38's unconditional proof with the positive kernel $|\psi_\Lambda|^2$ and $\gamma_\Lambda = \gamma_1 \Lambda^{-1/2}$ (Lemma 2.4, **[INTERNAL THEOREM]**). Consequently (Theorem 2.6, **[INTERNAL THEOREM]**) the Gromov–Hausdorff distance between the finite-first-moment state classes is at most γ_Λ. Unlike the Bochner–Riesz route, no Lebesgue constant enters, because the kernel is positive.
- **(Window choice, §2.4.)** The windows $\widehat{\psi_1} \propto (1-|\xi|^2)_+^m$ give $|\psi_1|^2 = c_m J_{m+1}(r)^2 / r^{2m+2}$, the squared modulus of the order-m Bochner–Riesz kernel. Its first moment is finite exactly when $m \geq 1$; $\gamma_1 = 32/(5\pi) \approx 2.04$ at $m = 1$ **[INTERNAL THEOREM + MEASURED]**. The sharp window ($m = 0$) has a logarithmically divergent first moment **[INTERNAL THEOREM + MEASURED]**.
- **(Why not Bochner–Riesz, §2.5.)** The two-dimensional Bochner–Riesz kernel is negative somewhere, and its Λ-independent Lebesgue constant exceeds 1, at every finite order **[PRIOR ART]** (classical [10]). The sizes are measured: $\min/\max = -0.071, -0.059, -0.029, -0.009$ at $s = 0.75, 1, 2, 4$ **[MEASURED]**.
- **(Finite disk, §3.)** On the Dirichlet disk the Markov–Cesàro map is unital. It is *not* positivity-preserving: a non-negative ring observable is sent to a function with negative values, at Cesàro orders 2 and 4. Its reconstruction error on the open disk decays on every tested observable. The operator system has propagation number two at the four truncations computed. All of these are **[MEASURED]**, except unitality, which is exact; no disk convergence theorem is claimed.

1.3 Honest scope

Withdrawn in the 2026-09-29 rebuild, each for the reason stated:

1. *Positivity of the two-dimensional Bochner–Riesz means.* False: the kernel oscillates (§2.5). The earlier test checked one positive Gaussian, which does not test positivity preservation.
2. *A "plane $\mathbb{R}^2_\alpha$" reconstruction and a backbone for the cigar-tip entropy of Paper 51.* The earlier plane map was defined through the order-0 Hankel transform, i.e. on the radial sector of the *flat* plane; the apex angle α never entered. The cone with $\alpha \neq 1$ has no translation group, so nothing in this paper reaches it (§4). Paper 51 now cites this paper only for the convergence proved here on the flat plane, leaving the disk and the cone open. Its convergence theorem for the discrete replica derivative (Theorem 3.1 there) does not depend on this paper.
3. *Plane van Suijlekom state-space convergence via a Bochner–Riesz tunnel.* Not established: its "height" leg carried a Lebesgue constant > 1 and the constituents did not match the framework (Remark 2.9). It is replaced by Theorem 2.6, proved by a different map.
4. *The finite-disk "non-decaying interior rate" $\Lambda^{+0.07}$ and the "Dirichlet-boundary obstruction".* The plateau near 0.3 came from evaluating the Markov ratio at the exact grid point $\rho = R$, where numerator and denominator are both floating-point rounding noise. On the open disk the error decays (§3). The boundary-collar error "≈ 0.26 at $R = 12$" is not reproduced: a direct recomputation gives $\sim 10^{-11}$.
5. *Cesàro positivity on the finite disk as a general property.* It held on the single tested observable $(\rho/R)^2$ and fails on a non-negative ring (§3).

Not claimed: any statement about Latrémolière's propinquity [4, 5, 7], any convergence on the finite disk or the cone, and novelty of the proof pattern (Remark 2.11).

2 The flat plane

2.1 Setup

Let $\mathcal{H} = L^2(\mathbb{R}^2)$, with unitary translations $(U_a\phi)(x) = \phi(x-a)$, $a \in \mathbb{R}^2$. For $\Lambda > 0$ let P_Λ be the Fourier projection onto $\mathcal{H}_\Lambda = \{\phi : \operatorname{supp}\widehat{\phi} \subset \{|\xi| \le \sqrt{\Lambda}\}\}$. Translations are Fourier multipliers, so $[U_a, P_\Lambda] = 0$.

Let $\mathcal{A} = C_{\mathrm{ub}}(\mathbb{R}^2)$, the bounded uniformly continuous functions (a unital C*-algebra on which $a \mapsto \tau_a f := f(\cdot - a)$ is norm-continuous), acting by multiplication operators M_f. Then $U_a M_f U_a^* = M_{\tau_a f}$. The truncated operator system is the norm closure

$$\mathcal{O}_\Lambda = \overline{\{S(f) := P_\Lambda M_f P_\Lambda \,:\, f \in \mathcal{A}\}} \subset B(\mathcal{H}_\Lambda),$$

unital since $S(1) = P_\Lambda$ is the identity of $B(\mathcal{H}_\Lambda)$. $S : \mathcal{A} \to \mathcal{O}_\Lambda$ is unital and completely positive.

Definition 2.1 (translation seminorms)**.** For $f \in \mathcal{A}$ and $T \in \mathcal{O}_\Lambda$,

$$L(f) = \sup_{a \neq 0} \frac{\|\tau_a f - f\|_\infty}{|a|}, \qquad L_\Lambda(T) = \sup_{a \neq 0} \frac{\|U_a T U_a^* - T\|_{\mathrm{op}}}{|a|}.$$

Since translations act transitively by isometries of the Euclidean metric, $L(f) = \mathrm{Lip}(f)$, and $L(f) = \|\nabla f\|_\infty$ for $f \in C^1$. On $\mathcal{H}_\Lambda$, $\|U_a - 1\|_{\mathrm{op}} \le |a|\sqrt{\Lambda}$ (the symbol $e^{-ia\cdot\xi}$ on the ball), so $L_\Lambda(T) \le 2\sqrt{\Lambda}\|T\|_{\mathrm{op}} < \infty$ for every T.

2.2 The window lift

Definition 2.2 (window and lift)**.** Let $\varphi \in L^2(\mathbb{R}^2)$ be supported in the closed unit ball, with $\|\varphi\|_2 = 1$, and let $\psi_1 = \mathcal{F}^{-1}\varphi$ (unitary Fourier transform). Assume

$$\gamma_1 := \int_{\mathbb{R}^2} |x|\,|\psi_1(x)|^2\,dx < \infty. \tag{1}$$

Set $\psi_\Lambda(x) = \Lambda^{1/2}\psi_1(\Lambda^{1/2}x)$, so that $\|\psi_\Lambda\|_2 = 1$ and $\widehat{\psi_\Lambda}$ is supported in $|\xi| \le \sqrt{\Lambda}$; hence $U_x\psi_\Lambda \in \mathcal{H}_\Lambda$ for every x. Write $k_\Lambda = |\psi_\Lambda|^2$, a probability density, and define

$$\upsilon(T)(x) = \langle U_x\psi_\Lambda,\, T\,U_x\psi_\Lambda\rangle, \qquad \gamma_\Lambda = \int |x|\,k_\Lambda(x)\,dx.$$

Lemma 2.3 (scaling; **[INTERNAL THEOREM]**)**.** $\gamma_\Lambda = \gamma_1\Lambda^{-1/2}$.

Proof. Substitute $y = \Lambda^{1/2}x$ in $\int |x|\,\Lambda|\psi_1(\Lambda^{1/2}x)|^2dx$. □

Lemma 2.4 (four estimates; **[INTERNAL THEOREM]**)**.** *For $f \in \mathcal{A}$ and $T \in \mathcal{O}_\Lambda$:*

(i) $L_\Lambda(S(f)) \le L(f)$;

(ii) $\upsilon(T) \in \mathcal{A}$, *$\upsilon$ is unital and positive, and* $L(\upsilon(T)) \le L_\Lambda(T)$;

(iii) $\upsilon(S(f))(x) = \int f(x+z)\,k_\Lambda(z)\,dz$, *hence* $\|\upsilon(S(f)) - f\|_\infty \le \gamma_\Lambda L(f)$;

(iv) $S(\upsilon(T)) = \Phi(T) := \int k_\Lambda(z)\,U_{-z}TU_{-z}^*\,dz$ *(weak integral), hence* $\|S(\upsilon(T)) - T\|_{\mathrm{op}} \le \gamma_\Lambda L_\Lambda(T)$.

Proof. (i) $U_a S(f) U_a^* - S(f) = P_\Lambda M_{\tau_a f - f} P_\Lambda$, and compression does not increase norms.

(ii) $|\upsilon(T)| \le \|T\|_{\mathrm{op}}$, and $\upsilon(1) = 1$, $\upsilon(T) \ge 0$ for $T \ge 0$, from the vector-state form. Since $U_x^* = U_{-x}$,

$$\upsilon(T)(x) - \upsilon(T)(x') = \big\langle \psi_\Lambda,\, U_{-x'}\big(U_{x'-x}TU_{x'-x}^* - T\big)U_{-x'}^*\,\psi_\Lambda\big\rangle,$$

whose modulus is at most $L_\Lambda(T)|x - x'|$. So $\upsilon(T)$ is bounded and Lipschitz, hence in $\mathcal{A}$, with $L(\upsilon(T)) \le L_\Lambda(T)$.

(iii) $U_x\psi_\Lambda \in \mathcal{H}_\Lambda$ absorbs the projections: $\upsilon(S(f))(x) = \int f(y)|\psi_\Lambda(y-x)|^2 dy = \int f(x+z)k_\Lambda(z)dz$. Then $|\upsilon(S(f))(x) - f(x)| \le \int |f(x+z)-f(x)|k_\Lambda(z)dz \le L(f)\gamma_\Lambda$.

(iv) For $T = S(f)$, (iii) gives $M_{\upsilon(T)} = \int k_\Lambda(z)\, M_{f(\cdot+z)}\, dz$ weakly, and $M_{f(\cdot+z)} = U_{-z}M_f U_{-z}^*$; compressing, and using $[U_{-z}, P_\Lambda] = 0$, gives $S(\upsilon(T)) = \Phi(T)$. Both sides are norm-contractive and linear in T, so the identity extends to the closure $\mathcal{O}_\Lambda$. Finally $\|\Phi(T)-T\|_{\mathrm{op}} \le \int k_\Lambda(z)\|U_{-z}TU_{-z}^* - T\|_{\mathrm{op}}dz \le L_\Lambda(T)\int |z|k_\Lambda(z)dz$. □

Test coverage. Facts (i), (ii) and (iv), and the identity half $\upsilon(S(f)) = f * k_\Lambda$ of (iii), are proof-only: no operator-level computation checks them. The numerical check (`tests/test_paper53_flat_plane_squa`) covers the convolution inequality $\|f * k_\Lambda - f\|_\infty \le \gamma_\Lambda L(f)$ only. It uses $f = |x - x_0|$, which is Lipschitz but unbounded and so lies outside C_{ub} (the inequality does not use boundedness, and there it is attained), and the bounded Lipschitz $f = \sin x_1$.

The positivity of k_Λ is used twice: in (iii) and (iv) the average of a quantity bounded by $L \cdot |z|$ is bounded by $L\gamma_\Lambda$. With a signed kernel of L^1 norm $\mathcal{L} > 1$ the same steps give no such bound (a signed kernel is not a state-induced average; compare Remark 2.9).

Corollary 2.5 (kernel; **[INTERNAL THEOREM]**). *$L_\Lambda(T) = 0$ if and only if $T \in \mathbb{C}1$.*

Proof. If $L_\Lambda(T) = 0$ then by (iv) $T = S(\upsilon(T))$, and by (ii) $\upsilon(T)$ is constant. □

2.3 State classes and the convergence statement

The Monge–Kantorovich (MK) metric on all states of $\mathcal{A}$ is not finite. For example $f_n(x) = \min(|x|, n)$ has $L(f_n) = 1$ and separates δ_0 from δ_x by $\min(|x|, n)$. Since $\mathcal{A}$ also has non-normal states at infinity, we restrict to states of finite first moment on both sides:

$$\mathcal{P}_1 = \{\mu : \text{Borel probability measure on } \mathbb{R}^2,\ \textstyle\int |x|\, d\mu < \infty\}, \qquad \mathcal{C}_\Lambda = \{\omega \in S(\mathcal{O}_\Lambda) :\ S^*\omega \in \mathcal{P}_1\},$$

where $S^*\omega = \omega \circ S$ and "$\in \mathcal{P}_1$" means that it is integration against such a measure. On $\mathcal{P}_1$ the MK metric of L is the Wasserstein distance W_1 (Kantorovich–Rubinstein duality [8]; bounded Lipschitz test functions suffice, by truncation, when first moments are finite). On $\mathcal{C}_\Lambda$ put $\mathrm{MK}_\Lambda(\omega, \omega') = \sup\{|\omega(T) - \omega'(T)| : T = T^* \in \mathcal{O}_\Lambda,\ L_\Lambda(T) \le 1\}$.

Theorem 2.6 (flat-plane state-space convergence). *Under Definition 2.2,*

$$d_{\mathrm{GH}}\Big((\mathcal{C}_\Lambda, \mathrm{MK}_\Lambda),\ (\mathcal{P}_1, W_1)\Big) \ \le\ \gamma_\Lambda = \gamma_1\Lambda^{-1/2} \ \xrightarrow[\Lambda\to\infty]{}\ 0.$$

Tiers. *The four estimates, the scaling and the kernel statement are **[INTERNAL THEOREM]** (Lemmas 2.3–2.4, Corollary 2.5). The Gromov–Hausdorff inequality is **[INTERNAL THEOREM]** for the two metric spaces named: the proof below closes on the classes $\mathcal{C}_\Lambda$, $\mathcal{P}_1$. We have not shown that these classes are the canonical ones (an open question, not a gap in the proof), and we have not connected them to Latrémolière's pointed/proper framework [7].*

Proof. Well-definedness. For $\mu \in \mathcal{P}_1$ put $\upsilon^*\mu(T) = \int \upsilon(T)\, d\mu$, a state of $\mathcal{O}_\Lambda$ by Lemma 2.4(ii). By (iii), $S^*\upsilon^*\mu$ is the law of $x + z$ with $x \sim \mu$ and $z \sim k_\Lambda$ independent, so its first moment is at most $\int |x| d\mu + \int |z| k_\Lambda(z) dz = \int |x| d\mu + \gamma_\Lambda < \infty$ (the triangle inequality; no symmetry of k_Λ is used); so $\upsilon^*\mu \in \mathcal{C}_\Lambda$. By definition S^* maps $\mathcal{C}_\Lambda$ into $\mathcal{P}_1$. MK_Λ is finite on $\mathcal{C}_\Lambda$: for $L_\Lambda(T) \le 1$, (iv) and (ii) give $|\omega(T) - \omega'(T)| \le W_1(S^*\omega, S^*\omega') + 2\gamma_\Lambda$. It separates states, since every self-adjoint T has $L_\Lambda(T) < \infty$.

Lipschitz pullbacks. By (i), $W_1(S^*\omega, S^*\omega') \le \mathrm{MK}_\Lambda(\omega, \omega')$; by (ii), $\mathrm{MK}_\Lambda(\upsilon^*\mu, \upsilon^*\nu) \le W_1(\mu, \nu)$.

Almost inverses. By (iii), $W_1(S^*\upsilon^*\mu, \mu) = \sup_{L(f)\le 1} |\int (\upsilon S f - f) d\mu| \le \gamma_\Lambda$; by (iv), $\mathrm{MK}_\Lambda(\upsilon^* S^*\omega, \omega) = \sup_{L_\Lambda(T)\le 1} |\omega(S\upsilon T - T)| \le \gamma_\Lambda$.

Correspondence. Let $\mathcal{R} = \{(\mu, \upsilon^*\mu)\} \cup \{(S^*\omega, \omega)\}$. For two pairs of the first kind, $\mathrm{MK}_\Lambda(\upsilon^*\mu, \upsilon^*\nu) \le W_1(\mu, \nu) \le W_1(S^*\upsilon^*\mu, S^*\upsilon^*\nu) + 2\gamma_\Lambda \le \mathrm{MK}_\Lambda(\upsilon^*\mu, \upsilon^*\nu) + 2\gamma_\Lambda$. For a mixed pair, $W_1(\mu, S^*\omega) \le$

$\mathrm{MK}_\Lambda(\upsilon^*\mu,\omega)+\gamma_\Lambda$ and $\mathrm{MK}_\Lambda(\upsilon^*\mu,\omega)\le W_1(\mu,S^*\omega)+\gamma_\Lambda$. The case of two pairs of the second kind is symmetric. The distortion is therefore at most $2\gamma_\Lambda$, and $d_{\mathrm{GH}}\le\gamma_\Lambda$. No compactness is used; both spaces are unbounded, and the bound is on the full (not pointed) Gromov–Hausdorff distance. □

2.4 Windows: smooth edge versus sharp edge

Proposition 2.7 (the $(1-|\xi|^2)^m$ windows; **[INTERNAL THEOREM + MEASURED]**). *Let $\varphi_m\propto(1-|\xi|^2)_+^m$, $m\in\{0,1,2,\dots\}$. By Sonine's integral [9],*

$$|\psi_1(x)|^2=c_m\,\frac{J_{m+1}(r)^2}{r^{2m+2}},\qquad r=|x|,\qquad c_m=\frac{4^m(m!)^2(2m+1)}{\pi},$$

which is the squared modulus of the two-dimensional Bochner–Riesz kernel of order m. It decays like r^{-2m-3}, so (1) holds if and only if $m\ge1$. For $m=1$, $\gamma_1=32/(5\pi)\approx2.037$; numerically $\gamma_1\approx2.299$ ($m=2$) and 2.593 ($m=3$).

The normalisation c_m and the $m=1$ moment follow from the Weber–Schafheitlin-type integral $\int_0^\infty J_\nu(t)^2t^{-\lambda}dt$. All three moments are reproduced by quadrature in `tests/test_paper53_flat_plane_squ` The π in γ_1 is the normalisation of the continuum Fourier–Bessel transform, a continuum-side constant. It is not a skeleton quantity, and no Paper 34 catalogue placement is claimed.

Remark 2.8 (the sharp window diverges)**. [INTERNAL THEOREM + MEASURED]** For $m=0$ (indicator of the ball), $k_1=J_1(r)^2/(\pi r^2)$. Its partial first moment is $\int_{|x|\le\rho}|x|k_1=2\int_0^\rho J_1(t)^2dt$, which grows like $(2/\pi)\log\rho$ by the Bessel asymptotics; the test measures the slope. So in two dimensions the sharp spectral window gives no finite rate, and a smooth-edged window is needed. On the compact group SU(2) the sharp (Fejér) window does give a finite first moment, $(4/\pi)\log n/n$ in Paper 38's normalisation. A natural reading is that the same slowly decaying tail is cut off there at the diameter. That reading is **[ARGUED]**; we have not checked it here.

2.5 Why the Bochner–Riesz map is not the lift

The earlier version of this paper used the two-dimensional Bochner–Riesz mean

$$\mathcal{B}^s_\Lambda(f)=\mathcal{F}^{-1}\big[(1-|\xi|^2/\Lambda)_+^s\,\widehat{f}\,\big]=K_{s,\Lambda}*f,\qquad K_{s,1}(r)\propto\frac{J_{s+1}(r)}{r^{s+1}},\tag{2}$$

as its reconstruction map. Two facts rule it out as a lift.

It does not preserve positivity. **[PRIOR ART]** (classical: the Bessel-kernel formula for the Fourier transform of a radial function [10], recorded here) J_{s+1} has infinitely many positive zeros and changes sign at each of them, so $K_{s,1}\propto J_{s+1}(r)/r^{s+1}$ is negative somewhere for every order s. The sizes are **[MEASURED]**: $\min K/\max K=-0.071,-0.059,-0.029,-0.009$ at $s=0.75,1,2,4$ (`tests/test_paper53_plane_bochner_riesz.py`). A non-negative f concentrated where $K<0$ is therefore mapped to a function with negative values. The positivity claimed in the earlier version ("classical, $s\ge\frac12$") is false. Bochner–Riesz summability above the critical index [10] concerns convergence, not positivity.

Remark 2.9 (the Lebesgue constant exceeds 1)**.** $\mathcal{B}^s_\Lambda$ is a convolution, so $\nabla(\mathcal{B}f)=\mathcal{B}(\nabla f)$, and by Young's inequality the sharp gradient gain is the Lebesgue constant $\mathcal{L}_s:=\|K_s\|_1/\int K_s$. This equals 1 only for a non-negative kernel. **[PRIOR ART]** (a classical fact about Bochner–Riesz means [10], recorded here) $\mathcal{L}_s>1$ at every finite order: $\int K_s=\widehat{K_s}(0)=1$ because the mean is unital, and K_s is negative on a set of positive measure (above), so $\|K_s\|_1>\int K_s=1$ (for $s\le\frac12$, $K_s\notin L^1$ and $\mathcal{L}_s=\infty$). **[MEASURED]** illustration, summing Bessel half-waves and

adding the analytic tail $\frac{2}{\pi}\sqrt{2/\pi}\,A^{1/2-s}/(s-\frac{1}{2})$ past the last zero A (stable to the digits shown from 200 to 3200 zeros):

s	0.6	0.75	1	1.5	2	3	5	8
$\mathcal{L}_s$	7.10	3.29	2.03	1.42	1.23	1.09	1.02	1.0045

$\mathcal{L}_s$ is Λ-independent by scaling and tends to 1 only as $s \to \infty$ (`tests/test_paper53_height_constant.py`). So $\mathcal{B}^s_\Lambda$ is not gradient-non-expansive over the unit-Lipschitz ball. This is not Paper 38's withdrawn Lemma L5 transported: L5's height was a reconstruction-defect quantity of a different construction.

Consequence for the construction. In the argument of Lemma 2.4, a lift of the form $T \mapsto K * (\text{something})$ with a signed kernel of L^1 norm $\mathcal{L}_s > 1$ is not a unital positive map. The "height" used in the earlier assembly (a Lipschitz gain) was also not Latrémolière's height, which is a state-space Hausdorff distance required to tend to zero along a convergent sequence [6, 7]. Both problems disappear with the squared window. Its kernel $|\psi_\Lambda|^2$, the squared modulus of a Bochner–Riesz kernel when $\widehat{\psi} \propto (1-|\xi|^2)^m_+$ (Proposition 2.7), is positive, so the constant is exactly 1 and the only rate is the first moment γ_Λ.

Remark 2.10 (the Bochner–Riesz rate on smooth observables)**. [MEASURED]** On individual smooth observables the Bochner–Riesz mean does reconstruct. For $f = e^{-\rho^2/2}$ at $s = 2$, $\|\mathcal{B}^s_\Lambda f - f\|_\infty / \|\nabla f\|_\infty$ decays roughly like $\Lambda^{-0.9}$ ($\Lambda = 5 \to 160$), and like $\Lambda^{-0.6}$ for a narrower Gaussian (`tests/test_paper53_plane_bochner_riesz.py`). These are observable-dependent rates. The uniform rate over the Lipschitz ball, which is what the Gromov–Hausdorff bound needs, is the first-moment rate $\gamma_\Lambda \propto \Lambda^{-1/2}$ of Theorem 2.6.

Remark 2.11 (prior art)**.** The template is Rieffel's [2]: for coadjoint orbits of compact Lie groups he pairs the Berezin covariant symbol (the coherent-state expectation, our υ) with its adjoint, the contravariant-symbol quantization, and bounds the Berezin transform by $\gamma = \int \rho(e,y)\, h_P(y)\, dy$, the first moment of the coherent-state transition probability $h_P(y) \propto |\langle U_y \xi, \xi\rangle|^2$ (his Theorem 2.3). Our γ_Λ is the same quantity for the translates of ψ_Λ; in place of his contravariant symbol we use the compression S. The compression/lifted-state pattern, with a single first-moment constant, is Gaudillot-Estrada and van Suijlekom's for compact metric groups [1]; Paper 38 [11] uses it on the S^3 spinor window. The present adaptation replaces the group by the non-compact translation group of $\mathbb{R}^2$, the Fejér kernel by a squared band-limited window, and the compact state spaces by finite-first-moment classes. We did not find a treatment of flat $\mathbb{R}^n$ spectral truncations of this kind in the literature we checked. This is a statement about our search, not a novelty claim.

3 The finite disk-with-cone: measurements

3.1 Setup

Let $\mathbb{D}_R$ be the flat disk of radius R with the Dirichlet Laplacian, eigenbasis

$$\psi_{k,j}(\rho,\phi) = N_{k,j}\, J_{|k|}\Big(\frac{a_{|k|,j}\rho}{R}\Big)\frac{e^{ik\phi}}{\sqrt{2\pi}}, \qquad \lambda_{k,j} = \Big(\frac{a_{|k|,j}}{R}\Big)^2, \tag{3}$$

with $a_{m,j}$ the j-th positive zero of J_m and $N_{k,j} = \sqrt{2}/(R\,|J_{|k|+1}(a_{|k|,j})|)$. On the disk-with-cone $\mathbb{D}_{R,\alpha}$ (azimuthal period $2\pi\alpha$) the half-integer fermionic azimuthal momentum is $m_{\text{eff}} = (k+\frac{1}{2})/\alpha$. Let P_Λ project onto $\lambda \le \Lambda$ and $\mathcal{O}_\Lambda = P_\Lambda C(\mathbb{D}_{R,\alpha}) P_\Lambda$.

Observation 3.1 (propagation number two; **[MEASURED]**)**.** In the truncated radial×azimuthal mode basis at $\alpha = 1$ (fermionic $m = |k+\frac{1}{2}|$), built from compressed monomials $z^a \bar{z}^b$, the squared operator system $\mathcal{O}^2$ spans the full envelope $M_N(\mathbb{C})$ at $(K,J) \in \{(2,2),(2,3),(3,2),(3,3)\}$, $N \in \{8,12,12,18\}$ (`tests/test_paper53_disk_obstruction.py`). This matches the Connes–van Suijlekom Toeplitz value [3]. It is not proved for all cutoffs.

3.2 The Markov–Cesàro map

Definition 3.2 (Markov–Cesàro reconstruction)**.** For order s and cutoff Λ, with $w_a = (1 - \lambda_a/\Lambda)_+^s$,

$$g_\Lambda(x) = \frac{\sum_a w_a \langle \psi_a, f\rangle \psi_a(x)}{\sum_a w_a \langle \psi_a, 1\rangle \psi_a(x)}, \qquad \mathcal{B}_\Lambda(f) = P_\Lambda M_{g_\Lambda} P_\Lambda. \tag{4}$$

Proposition 3.3 (what holds and what does not; **[MEASURED]** except (a))**.** *In the radial ($k = 0$) sector, $R = 1$, $\Lambda = 100$ to 1600 (and 6400 where stated):*

(a) Unitality *(exact):* $f \equiv 1 \Rightarrow g_\Lambda \equiv 1$.

(b) Positivity fails in general. *For a non-negative ring $f(\rho) = e^{-((\rho-\rho_0)/0.02)^2}$, centred where the Cesàro kernel is most negative,* $\min g_\Lambda = -0.016, -0.025, -0.045$ *at* $\Lambda = 100, 400, 1600$ *($s = 2$). At $s = 4$ it is -0.006 and -0.011 at $\Lambda = 400, 1600$. The denominator is positive at those points. The interior Cesàro kernel has* $\min / \max \approx -0.03$ *($s = 2$) and -0.009 ($s = 4$), matching the plane Bochner–Riesz values of §2.5. The failure is also operator-level: the compressed operator $\mathcal{B}_\Lambda(f) = P_\Lambda M_{g_\Lambda} P_\Lambda$, restricted to the $k = 0$ sector, has smallest eigenvalue $-0.0034, -0.0022, -0.0022$ at $\Lambda = 100, 400, 1600$ ($s = 2$) and -0.00044 at $\Lambda = 1600$ ($s = 4$; at $s = 4$, $\Lambda = 400$ it is positive, $+0.0015$), so $\mathcal{B}_\Lambda$ sends a positive f to a non-positive operator. On the single observable $(\rho/R)^2$,* $\min g_\Lambda > 0$ *at $s = 2$; this is where the earlier general positivity claim came from. The truncated heat weight $e^{-\lambda/\Lambda}$ already fails on $(\rho/R)^2$ (*$\min g \approx -0.13$ *at $\Lambda = 200$).*

(c) The reconstruction error on the open disk decays. *For $f = (\rho/R)^2$, which is non-zero at the boundary,* $\sup_{\rho<R} |g_\Lambda - f|/\|\nabla f\|_\infty = 0.237, 0.179, 0.133, 0.097, 0.070$ *at* $\Lambda = 100, 200, 400, 800, 1600$ *(about $\Lambda^{-0.44}$). The maximum sits at the boundary. For an interior-supported C^1 bump the error falls roughly like $\Lambda^{-0.9}$ (*$0.163 \to 0.0032$ *over* $\Lambda = 100 \to 6400$*).*

Remark 3.4 (the withdrawn "Dirichlet-boundary obstruction")**.** Earlier versions reported that the disk error does not decay ($\sim \Lambda^{+0.07}$, plateau ≈ 0.3) and read this as an obstruction intrinsic to the Dirichlet boundary. The plateau came from the single grid point $\rho = R$. There every $J_0(a_j)$ is rounding noise of order 10^{-16}, so the ratio (4) returns an arbitrary number. Dropping that one point, or evaluating at $R - 10^{-9}$, gives the decaying values of Proposition 3.3(c) (the test now pins both). The companion claim that the boundary-collar error for $f = e^{-\rho^2/2}$ on $R = 12$ stays near 0.26 is not reproduced: a direct recomputation gives $\sim 10^{-11}$, in line with the sprint driver's own docstring. What the boundary does cause is a slower rate for observables that are non-zero there. The disk measurements therefore do *not* show an obstruction. They show that the obvious reconstruction map fails positivity, so no proof route is available from this map.

Remark 3.5 (why no disk theorem)**.** The argument of §2 needs a transitive group of isometries commuting with the truncation, and a positive lift. The disk has neither: its isometries are the rotations about the centre, and the Markov–Cesàro map is not positive. Whether the disk truncations converge in any state-space sense is open (§5, Q2).

4 The cone $\alpha \neq 1$

The cone $\mathbb{R}^2_\alpha$ (azimuthal period $2\pi\alpha$) is flat away from its apex, but for $\alpha \neq 1$ it is not homogeneous. Its only isometries are rotations about the apex, so there is no translation seminorm and no family of window translates to average over. The earlier "plane $\mathbb{R}^2_\alpha$" construction never reached the cone: it used the order-0 Hankel transform, the radial sector of the flat plane, and α did not appear. Convergence of spectral truncations of the cone, and in particular anything bearing on the apex-local cigar-tip entropy of Paper 51, is open (Q1).

5 Open questions

Q1 (the cone). Is there a state-space convergence statement for band-limited (Hankel-order $|k+\frac{1}{2}|/\alpha$) truncations of the cone $\mathbb{R}^2_\alpha$? A lift would have to be built without translations. One option is window vectors localised along geodesics, with a separate estimate near the apex.

Q2 (the finite disk). Does a positive, unital reconstruction map with a decaying Lipschitz-uniform rate exist for the Dirichlet disk? Such a map would, for instance, be built from localised vectors in the truncated space rather than from Cesàro weights. And is the propagation number two for all cutoffs (Observation 3.1)?

Q3 (optimal window and sharp constant). Among admissible windows the rate is $\gamma_1\Lambda^{-1/2}$. Which window minimises γ_1, and is $\Lambda^{-1/2}$ the sharp order of the Gromov–Hausdorff distance, or only of this bound?

Q4 (Latrémolière's propinquity). Theorem 2.6 is a Gromov–Hausdorff statement on chosen state classes. Whether the same pair (S, v) yields convergence in the pointed/proper propinquity [7] is open.

Q5 ($\mathbb{R}^n$). The argument is dimension-independent in form, with $\gamma_\Lambda = \gamma_1\Lambda^{-1/2}$ in every dimension. Which windows have a finite first moment in dimension n was not checked.

6 Conclusion

On the flat plane, spectral truncation converges in the state-space Gromov–Hausdorff sense at rate $\gamma_1\Lambda^{-1/2}$, provided the lift uses a window whose squared modulus has a finite first moment. The four estimates behind this are proved, and so is the final Gromov–Hausdorff step, on the finite-first-moment state classes we chose (their canonicity is open). The Bochner–Riesz map is not a usable lift, because its kernel changes sign. Replacing it by the squared modulus of such a kernel removes the Lebesgue constant. On the finite disk-with-cone the earlier obstruction was a numerical artifact, and the obvious reconstruction map is not positive. The disk and the cone with $\alpha \neq 1$ are open.

Acknowledgments

The finite-disk measurements and the Lebesgue-constant table were made in the GeoVac repository and are frozen in the `tests/test_paper53_*` files cited above. The 2026-09-29 rebuild, including the retractions in §1.3, is recorded in the CHANGELOG.

References

[1] Y. Gaudillot-Estrada and W. D. van Suijlekom, "Convergence of spectral truncations for compact metric groups," *Int. Math. Res. Not. IMRN* **2025**, no. 13 (2025), rnaf197; arXiv:2310.14733.

[2] M. A. Rieffel, "Matrix algebras converge to the sphere for quantum Gromov–Hausdorff distance," *Mem. Amer. Math. Soc.* **168** (2004), no. 796, 67–91; arXiv:math/0108005.

[3] A. Connes and W. D. van Suijlekom, "Spectral truncations in noncommutative geometry and operator systems," *Comm. Math. Phys.* **383** (2021), 2021–2067.

[4] F. Latrémolière, "The dual Gromov–Hausdorff propinquity," *J. Math. Pures Appl.* **103** (2015), 303–351.

[5] F. Latrémolière, "Quantum metric spaces and the Gromov–Hausdorff propinquity," *Contemp. Math.* **676** (2016), 47–133.

[6] F. Latrémolière, "The Gromov–Hausdorff propinquity for metric spectral triples," *Adv. Math.* **404** (2022), Paper No. 108393, 56 pp.; preprint arXiv:1811.10843.

[7] F. Latrémolière, "The quantum Gromov–Hausdorff hypertopology on the class of pointed proper quantum metric spaces," arXiv:2512.03573 (2025).

[8] C. Villani, *Optimal Transport: Old and New*, Grundlehren der mathematischen Wissenschaften **338**, Springer, 2009 (Kantorovich–Rubinstein duality).

[9] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge Univ. Press, 1944 (Sonine's finite integral; Weber–Schafheitlin integrals).

[10] E. M. Stein and G. Weiss, *Introduction to Fourier Analysis on Euclidean Spaces*, Princeton Univ. Press, 1971, Ch. IV (Fourier transforms of radial functions; the Bessel-kernel formula) and Ch. VII (Bochner–Riesz summability).

[11] GeoVac Project. Paper 38 — State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3. `papers/group1_operator_algebras/paper_38_su2_propinquity_convergence.tex`.

[12] GeoVac Project. Paper 40 — State-space Gromov–Hausdorff convergence of spectral truncations of compact semisimple Lie groups with bi-invariant metric. `papers/group1_operator_algebras/paper_40_unified_propinquity_convergence.tex`.

[13] GeoVac Project. Paper 51 — A Convergent Discrete Replica Derivative on the Spinor Cone, with the Two-Term Camporesi–Higuchi Spectral Action on $S^3 \times S^1$. `papers/group5_qed_gauge/paper_51_gravity_arc.tex`.